

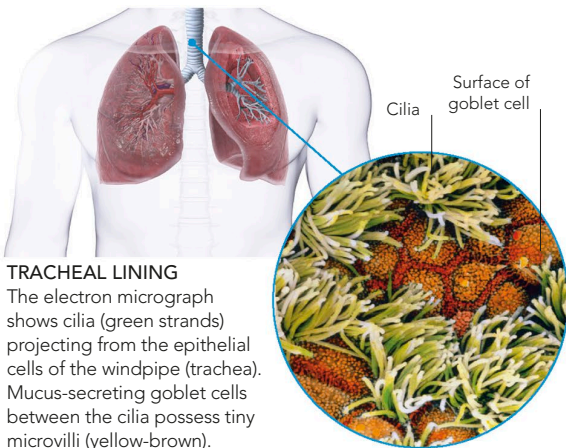
EPITHELIUM

Epithelial tissue, also called epithelium, is an important structural element that acts as a lining or covering for other body tissues. Epithelium can be classified according to the shape and layout pattern of individual cells (see opposite), and also by the arrangement of cells into one or more layers. Most epithelial tissues form

membranes and are specialized for protection, absorption, or secretion. They do not contain blood vessels, and their cells are usually anchored to, and stabilized by, a basement membrane. There may be other cell types present, such as goblet cells that secrete blobs of mucus for release on to the surface.

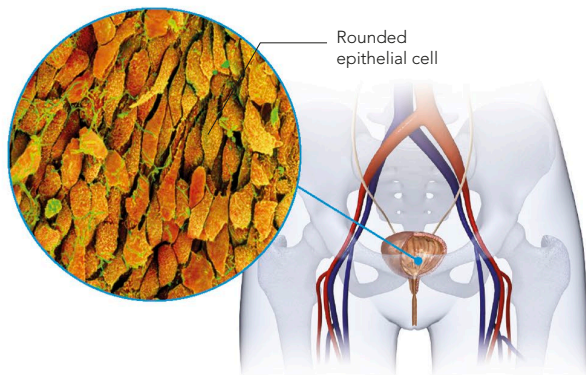
PSEUDOSTRATIFIED EPITHELIUM

This type of columnar epithelium seems to be arranged in vertical layers. However, it actually consists of a single layer of cells of varying shapes and heights. The nuclei (control centres) of the different cell types are also at different levels, creating a layered (stratified) effect. Taller cells may be specialized into mucus-making goblet cells or ciliated cells that trap foreign particles. This type of epithelium occurs in the airway linings, and in the excretory and male reproductive passages and ducts.



TRACHEAL LINING

The electron micrograph shows cilia (green strands) projecting from the epithelial cells of the windpipe (trachea). Mucus-secreting goblet cells between the cilia possess tiny microvilli (yellow-brown).



TRANSITIONAL EPITHELIUM

This epithelial tissue is similar to layered (stratified) epithelium, but is able to stretch without tearing. There are usually columnar cells in the basal layer, which become more rounded in the upper layers. As these layers stretch, the cells flatten, or become more squamous. Transitional epithelium is well suited to the urinary system, where it lines areas within the kidneys, ureters, bladder, and urethra. It allows these organs to bulge as urine flows through at pressure. The epithelium also secretes mucus that protects it from acidic urine.

BLADDER LINING

The electron micrograph shows the tightly packed epithelial cells of the bladder lining. They are soft and pliable, enabling them to stretch as the bladder fills with urine.